University of Central Punjab

Faculty of Information Technology

Object Oriented Programming

Fall 2025

Lab 08	
Topic	Classes in C++, access specifiers, objects, methods, constructors, association, aggregation, composition
Objective	Making students familiarize with the concepts of constructors, association, aggregation, composition
CLO	Apply object-oriented programming principles to implement real-world problems.

Instructions:

- Indent your code.
- Comment your code.
- Use meaningful variable names.
- Plan your code carefully on a piece of paper before you implement it. •
  Name of the program should be same as the task name. i.e. the first program should be Task_1.cpp

- **void main() is not allowed. Use int main()**
- **You are not allowed to use any built-in functions**
- **You are required to follow the naming conventions as follow:**
- **Variables:** firstName; (no underscores allowed)
- **Function:** getName(); (no underscores allowed)
- **ClassName:** BankAccount (no underscores allowed)

Students are required to complete the following tasks in lab timings.

Task 1: Association (One-to-One, One-to-Many, Many-to-Many)

Objective:

Design two classes Person and Passport. Each person has exactly one passport

Write C++ class definitions using private attributes and appropriate setters/ getters

For passport class the attributes are:

- int passportNumber
- string nationality

For person class the attributes are:

- Passport* passport
- string name

Required Functions:

- **Parameterized Constructor**:
- **Setter and Getter functions**:
- **Display function**

Main Function:

In the main function, a Passport object is created using a passport number and nationality. Then, a Person object is created by passing the person's name and the address of the Passport object.
Finally, information such as the person's name, passport number, and nationality is accessed and displayed.

Task 2: One-to-Many Association

Objective:

Create a class Teacher and a class Course. A teacher can teach multiple courses.

Write the C++ class declarations using an array or pointer-based structure.

For class Course, the attributes are:

- string courseCode
- string title

For class Teacher, the attributes are:

- Course*courses[5]
- string name
- int courseCount

Required Functions:

- **Parameterized Constructor**.
- **Implement Getters and Setters**
- **Member Function**:
  - o Implement a display function

Main Function:

In the main function, a Teacher object is created using the teacher's name. Then, multiple Course objects are created with course codes and titles. These courses are added to the teacher using the addCourse method. Finally, the course codes of the teacher's courses are retrieved and displayed.

Task 3: Many-to-Many Association

Objective:

Design classes Student and Club. A student can join multiple clubs, and a club can have multiple students.

Create a link class Membership.

For class Course, the attributes are:

- int studentID
- string name

For class Club, the attributes are:

- string clubname

For class Membership, the attributes are:

- Student* student
- Club* club

Required Functions:

- **Parameterized Constructor**
- **Setter and Getter Methods**
- **Display Function**

Main Function:

In the main function, a Student object is created with a student ID and name. A Club object is also created with a club name. A Membership object is then created by associating the student with the club, establishing their membership relationship.

Task 4: Aggregation

Objective:

Model a Library and Book. A library has books but books can exist independently.

Represent this using aggregation and implement the class headers.

For class Book, the attributes are:

- string title
- string author

For class Library, the attributes are:

- Books* books[10]

- int bookCount

Required Functions:

- **Parameterized Constructor**
- **Getters and Setters**
- **Display Functions**
- **Member Function**
  (addBook**())**

Main Function:

In the main function, a Library object is created to manage a collection of books. Multiple Book objects are created using titles and author names. These books are then added to the library using the addBook method.

Task 5: Composition

Design classes Car and Engine. Every car must have an engine, and if the car is destroyed, so is the engine.

For class Engine, the attributes are:

- string type

- int horsepower

For class Car, the attributes are:

- Engine* engine

Required Functions:

- **Default Constructor**
- **Parameterized Constructor**
- **Setter and Getter Methods**
- **Display Function**

Main Function:

In the main function, a Car object is created by specifying the engine type and horsepower. The Engine is constructed internally within the Car object using composition, meaning it is a built-in part of the car rather than being passed from outside..